

CLASS TEST - 2

① Using ~~Maxwell's~~ Same as

A6/83

② Show that the general solution of the 1+1 dimensional wave eqⁿ

$$\frac{1}{v_0^2} \frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = 0$$

can be expressed as $f(t, x) = f_+(v_0 t + x) + f_-(v_0 t - x)$

Ans: C ~~Let~~ $u_+ = v_0 t + x$
 $u_- = v_0 t - x$

$$f = f_+(u_+) + f_-(u_-)$$

$$\rightarrow \frac{\partial f}{\partial t} = \frac{\partial f_+}{\partial u_+} \frac{\partial u_+}{\partial t} + \frac{\partial f_-}{\partial u_-} \frac{\partial u_-}{\partial t}$$

$$\rightarrow \frac{\partial f}{\partial x} = \frac{\partial f_+}{\partial u_+} \frac{\partial u_+}{\partial x} + \frac{\partial f_-}{\partial u_-} \frac{\partial u_-}{\partial x}$$

$$\frac{\partial^2 f}{\partial t^2} = v_0^2 \left[\frac{\partial^2 f_+}{\partial u_+^2} + \frac{\partial^2 f_-}{\partial u_-^2} \right]$$

$$\frac{\partial^2 f}{\partial x^2} = \left[\frac{\partial^2 f_+}{\partial u_+^2} + \frac{\partial^2 f_-}{\partial u_-^2} \right]$$

$$\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = 0 \quad \text{--- ①}$$

$$\therefore \frac{\partial f}{\partial x} = \frac{\partial f_+}{\partial u_+} \cdot \frac{\partial u_+}{\partial x} + \frac{\partial f_-}{\partial u_-} \cdot \frac{\partial u_-}{\partial x}$$

$$\rightarrow \boxed{\frac{\partial f}{\partial x} = \frac{\partial f_+}{\partial u_+} - \frac{\partial f_-}{\partial u_-}}$$

$$\therefore \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f_+}{\partial u_+^2} \cdot \frac{\partial u_+}{\partial x} - \frac{\partial^2 f_-}{\partial u_-^2} \cdot \frac{\partial u_-}{\partial x}$$

$$\rightarrow \boxed{\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f_+}{\partial u_+^2} + \frac{\partial^2 f_-}{\partial u_-^2}} \quad \text{--- (ii)}$$

From (i) & (ii), it can be seen that :

$$\boxed{\frac{1}{v_0^2} \frac{\partial^2 f}{\partial t^2} = \frac{\partial^2 f}{\partial x^2}} \quad \text{(proved)}$$

Hence, the general solⁿ can be expressed as $f(t, x)$.